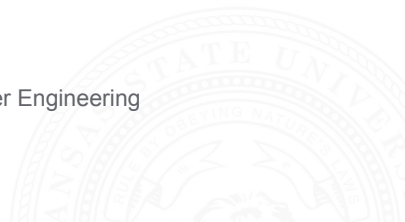


Single-Phase AC Circuits III

Power-factor correction with shunt capacitors

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Lecture outline

- 1 Calculate capacitor compensation for a target power factor
- 2 Distinguish absorbed and supplied reactive power
- 3 Verify the corrected power factor and source current

PART 1

Theory and formulas

Power conventions and the four-step correction method

Load and generator sign conventions

Recall L06: L = load, G = generator

Load: power absorbed

I_L enters the + terminal.

$$S_L = V_L I_L^* = P_L + jQ_L$$

- $P_L > 0$: absorbs real power.
- $Q_L > 0$: absorbs reactive power.
- $Q_L < 0$: supplies reactive power.

Generator: power delivered

I_G leaves the + terminal.

$$S_G = V_G I_G^* = P_G + jQ_G$$

- $P_G > 0$: supplies real power.
- $Q_G > 0$: supplies reactive power.
- $Q_G < 0$: absorbs reactive power.

Positive means absorption at the load and delivery at the generator.

Capacitor power and power balance

The capacitor uses the load convention

I_C enters the + terminal. An ideal capacitor has $P_C = 0$.

$$S_C = VI_C^* = jQ_C, \quad Q_C < 0$$

The source supplies the load and capacitor together

$$S_G = S_L + S_C$$

$$P_G = P_L, \quad Q_G = Q_L + Q_C$$

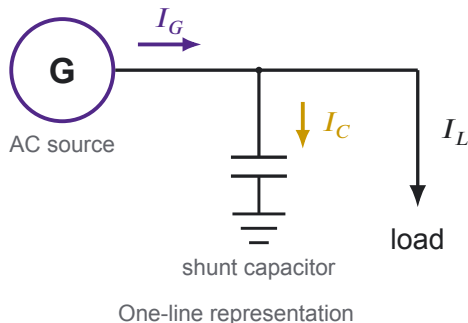
Negative Q_C means reactive power supply. It reduces Q_G while Q_L stays unchanged.

Shunt compensation

What remains fixed

- The load voltage is maintained at the declared value.
- The original load still absorbs $P_L + jQ_L$.
- An ideal capacitor has zero average real-power absorption.

The capacitor supplies part of the load's reactive power at the load terminals.



Power-factor correction

Known load $P = P_L$, $Q_1 = Q_L$; **target source reactive power** Q_2

Fixed RMS voltage V . The capacitor uses $Q_C < 0$.

- 1 Target reactive power: $Q_2 = P \tan(\cos^{-1} \text{pf}_2)$.
- 2 Required compensation: $Q_C = Q_2 - Q_1$.
- 3 Capacitance: $C = |Q_C|/(2\pi f V^2)$.
- 4 Verification: check power balance, the target power factor, and reduced source current.

Step 1: Target reactive power

Known load: P and $Q_1 = Q_L > 0$

For a target lagging power factor pf_2 ,

$$\phi_2 = \cos^{-1}(\text{pf}_2), \quad \tan \phi_2 = \frac{Q_2}{P}.$$

$$Q_2 = P \tan(\cos^{-1}(\text{pf}_2))$$

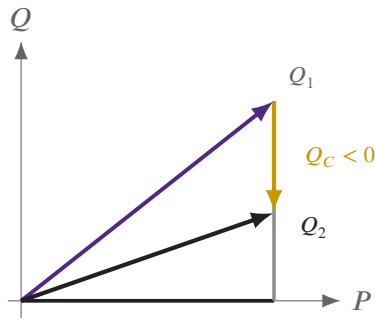
Q_2 is the target source reactive power. The original load still absorbs Q_1 .

Step 2: Required compensation

$$Q_C = Q_2 - Q_1 < 0$$

$$S_C = jQ_C, \quad \text{Im}\{S_C\} < 0$$

$Q_C < 0$ follows the load convention. The capacitor supplies $-Q_C$ kvar.



Step 3: Capacitance

For a shunt capacitor across RMS voltage V ,

$$|Q_C| = \omega C V^2, \quad \omega = 2\pi f.$$

$$C = \frac{|Q_C|}{\omega V^2} = \frac{|Q_C|}{2\pi f V^2}$$

Voltage V_{rms} **across the capacitor**

Use the voltage measured directly across the capacitor terminals.

Units $\text{var}/[(\text{rad/s})V^2] = \text{F}$

Convert kvar to var before substitution.

Step 4: Verification

Use the selected capacitor to recover the corrected source state

$$S_G = S_L + S_C = P + j(Q_1 + Q_C)$$

$$|S_G| = \sqrt{P^2 + (Q_1 + Q_C)^2}, \quad \text{pf}_{\text{check}} = \frac{P}{|S_G|}$$

$$I_G = \frac{S_G^*}{V}, \quad |I_G| = \frac{|S_G|}{V} \quad (V \text{ as phase reference}).$$

Confirm unchanged P , the target power factor, and smaller source current.

PART 2

Worked example

The L06 load: correction from 0.800 to 0.950 lagging

Worked example: the L06 load

Known load and results from L06

$$V = 240\angle 0^\circ \text{ V}_{\text{rms}}, \quad Z_L = 8 + j6 \, \Omega$$

$$P = 4.608 \text{ kW}, \quad Q_1 = 3.456 \text{ kvar}, \quad \text{pf}_1 = 0.800 \text{ lagging}$$

$$|S_{G1}| = 5.760 \text{ kVA}, \quad |I_{G1}| = 24.0 \text{ A}$$

Correction target and capacitor conditions

$$\text{pf}_2 = 0.950 \text{ lagging}, \quad f = 60 \text{ Hz.}$$

Apply Steps 1–4 to this same load using its known initial state.

Example: Step 1

Target reactive power

$$\phi_2 = \cos^{-1}(0.950) = 18.195^\circ$$

$$Q_2 = P_L \tan \phi_2 = (4.608)(0.328684) \approx \boxed{1.515 \text{ kvar}}.$$

$Q_2 = 1.515 \text{ kvar}$ is the source target. Keep $Q_1 = 3.456 \text{ kvar}$ for the load.

Example: Step 2

Required compensation

$$Q_C = Q_2 - Q_1 = 1.515 - 3.456 = \boxed{-1.941 \text{ kvar}}.$$

$$\boxed{S_C = -j1.941 \text{ kVA}}$$

Rating language 1.941 kvar **supplied**

A positive equipment-rating magnitude.

Power check $Q_C = -1.941 \text{ kvar}$

Negative under the passive convention.

Example: Step 3

Capacitance

$$C = \frac{|Q_C|}{2\pi f V^2} = \frac{1941.424}{2\pi(60)(240)^2} = 8.941 \times 10^{-5} \text{ F}$$

$$C = 89.4 \mu\text{F}$$

Back-substitution

The capacitor reactive-power magnitude is recovered.

$$2\pi(60)(89.4 \mu\text{F})(240)^2 \approx 1.941 \text{ kvar}$$

Example: Step 4

Verification

Quantity	Before	After	Interpretation
Load P	4.608 kW	4.608 kW	unchanged load demand
Source Q_G	3.456 kvar	1.515 kvar	capacitor supplies the difference
$ S_G $	5.760 kVA	4.851 kVA	15.8% reduction
$ I_G $ at 240 V	24.00 A	20.21 A	same 15.8% reduction

$$S_G = S_L + S_C = 4.608 + j[3.456 + (-1.941)] = 4.608 + j1.515 \text{ kVA}$$

$\text{pf}_{\text{check}} = 4.608/4.851 \approx 0.950$ **lagging. Source current falls from 24.00 to 20.21 A.**

Correction summary

- 1 Target reactive power: $Q_2 = P \tan(\cos^{-1} \text{pf}_2)$.
- 2 Required compensation: $Q_C = Q_2 - Q_1$.
- 3 Capacitance: $C = |Q_C|/(2\pi f V^2)$ at the capacitor terminals.
- 4 Verification: check net source power, power factor, and current.

At fixed voltage, load power stays unchanged. The capacitor supplies part of the reactive demand.